

Cost-Effectiveness Analysis Report

Biomarker-Guided Strategies for Metastatic MSS/pMMR Colorectal Cancer

Ben Geisler

2025-10-06

Table of contents

1	Executive Summary	2
2	Introduction	2
2.1	Study Objectives	2
3	Methods	2
3.1	Model Structure and Approach	2
3.2	Biomarker Strategies	2
3.3	Quality of Life Analysis	3
3.4	Survival Analysis	3
3.5	Model Parameters	3
4	Results	3
4.1	Base Case Analysis	3
4.2	Sensitivity Analyses	5
4.2.1	Deterministic Sensitivity Analysis	5
4.2.2	Probabilistic Sensitivity Analysis	6
5	Discussion	7
5.1	Key Findings	7
5.2	Clinical Implications	8
5.3	Economic Implications	8
5.4	Limitations	8
6	Conclusions	8

1 Executive Summary

This cost-effectiveness analysis evaluates three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients for immunotherapy versus the standard of care strategy of chemotherapy alone. The analysis employs a comprehensive decision-analytic model incorporating clinical trial data, parametric survival analysis, and uncertainty quantification through deterministic and probabilistic sensitivity analyses.

2 Introduction

Metastatic colorectal cancer represents a significant clinical and economic burden. This analysis compares three biomarker-guided approaches against standard chemotherapy to inform treatment decisions and healthcare resource allocation. The biomarker strategies evaluate different approaches to patient selection for immunotherapy based on C-reactive protein (CRP), tumor lesion reduction (TLR), and combined tumor mutational burden/BRAF mutation status (TMB/BRAF).

2.1 Study Objectives

1. Evaluate the cost-effectiveness of CRP, TLR, and TMB/BRAF biomarker strategies
2. Compare population-level outcomes across all strategies
3. Assess the impact of biomarker prevalence on overall cost-effectiveness
4. Provide uncertainty analysis through deterministic and probabilistic sensitivity analyses

3 Methods

3.1 Model Structure and Approach

The analysis employs a decision-analytic model structure with a 10-year time horizon and weekly cycle length. The model incorporates appropriate discount rates for costs and health outcomes and evaluates strategies from a healthcare payer perspective.

3.2 Biomarker Strategies

Three biomarker strategies are compared against standard care:

1. **CRP Strategy:** C-reactive protein biomarker guidance with <5 mg/L cutoff for biomarker-positive status
2. **TLR Strategy:** Tumor lesion reduction biomarker guidance with 10% reduction cutoff
3. **TMB/BRAF Strategy:** Combined tumor mutational burden (9 mutations/Mb) or BRAF V600 mutation guidance

For biomarker-positive patients, treatment consists of alternating cycles of FLOX chemotherapy and nivolumab immunotherapy. Biomarker-negative patients receive standard FLOX chemotherapy only.

3.3 Quality of Life Analysis

Quality-adjusted life years were derived from patient-reported outcome measures using EQ-5D-5L data where available. Health state utilities were estimated for progression-free and progressed disease states.

3.4 Survival Analysis

Parametric survival models were fitted to overall survival and progression-free survival endpoints. Multiple distributions were tested, with model selection based on Akaike Information Criterion (AIC). The analysis included age, sex, treatment assignment, and biomarker interaction terms.

3.5 Model Parameters

The model incorporates clinical parameters from the METIMMOX trial, cost parameters based on German healthcare tariffs, and utility parameters derived from EQ-5D-5L questionnaires. All parameters include appropriate uncertainty distributions for probabilistic sensitivity analysis.

4 Results

4.1 Base Case Analysis

Table 1: Base Case Cost-Effectiveness Results with ICERs

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,827	1.892	NA	NA	NA	ND
crp	\$56,299	2.014	34471.96	0.121884	\$282,826	ND
tmb_braf	\$68,664	1.944	NA	NA	NA	D
tlr	\$77,574	1.359	NA	NA	NA	D

The base case analysis presents the incremental cost-effectiveness ratios (ICERs) for each biomarker strategy compared to the reference strategy. Results include total costs, quality-adjusted life years (QALYs), and incremental cost-effectiveness ratios with efficiency frontier analysis.

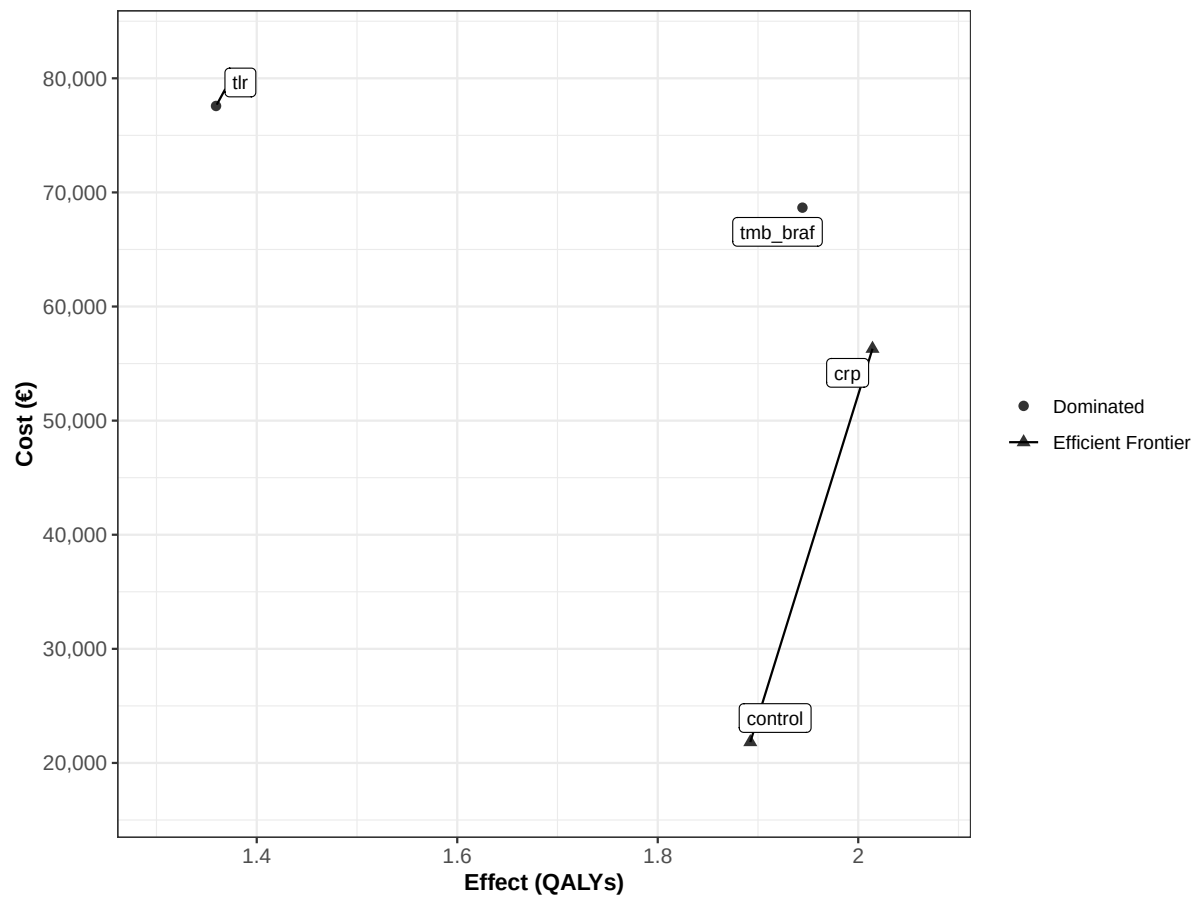


Figure 1: Cost-Effectiveness Plane with Efficiency Frontier

The cost-effectiveness plane displays the relationship between costs and health outcomes for all strategies, with the efficiency frontier indicating the optimal strategies at different willingness-to-pay thresholds.

4.2 Sensitivity Analyses

4.2.1 Deterministic Sensitivity Analysis

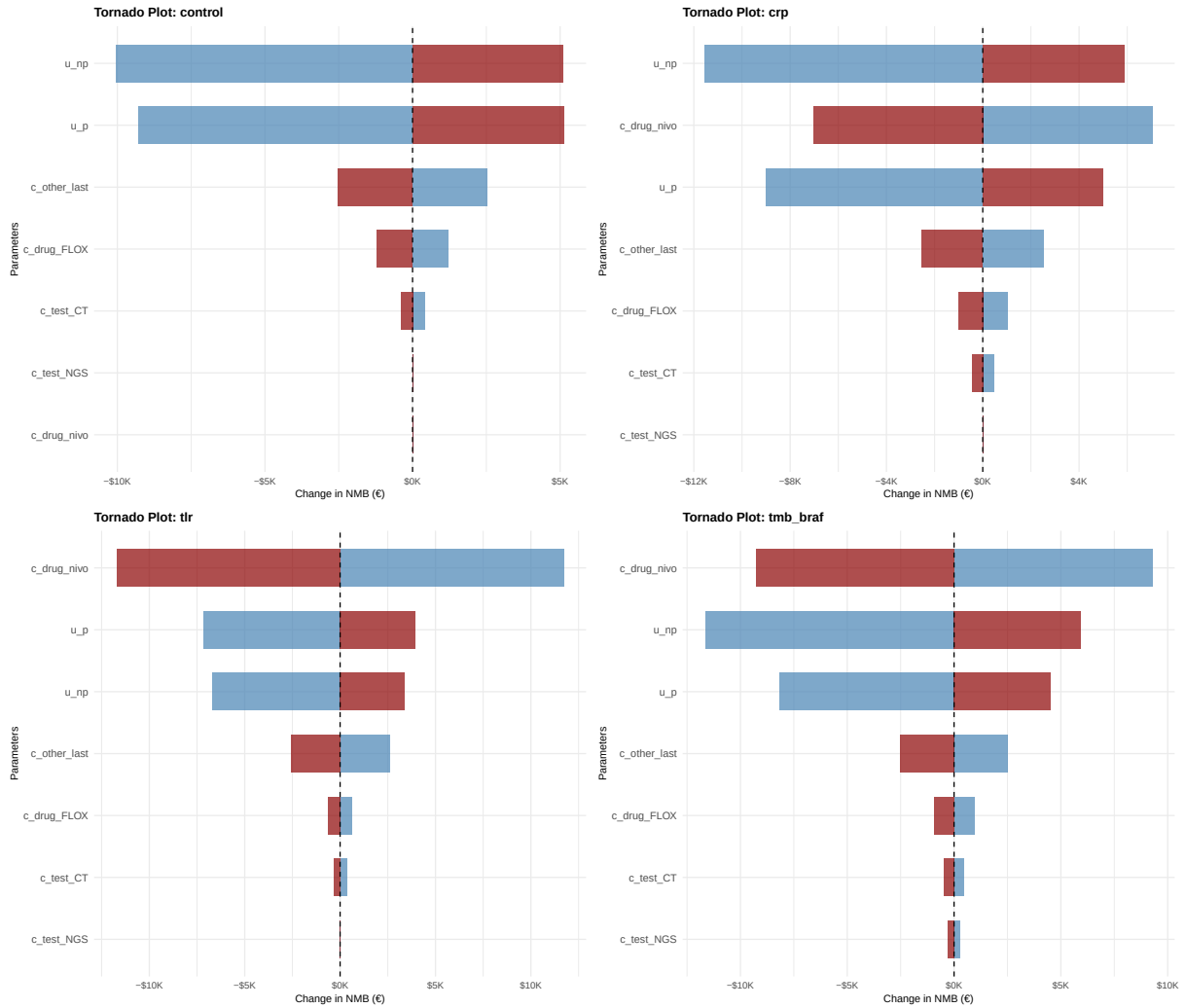


Figure 2: Tornado Diagrams - One-Way Sensitivity Analysis for All Strategies

****Tornado diagrams generated for 4 strategies.****

Table 2: Parameter Impact Summary - Top 3 Most Influential Parameters by Strategy

Strategy	1st Most Influential		2nd Most Influential		3rd Most Influential	
	Parameter	Impact	Parameter	Impact	Parameter	Impact
Standard Care	Utility (No Progression)	\$15,131	Utility (Progression)	\$14,392	End-of-Life Care Cost	\$5,131
CRP-guided	Utility (No Progression)	\$17,401	Nivolumab Cost	\$14,084	Utility (Progression)	\$13,401
TLR-guided	Nivolumab Cost	\$23,444	Utility (Progression)	\$11,127	Utility (No Progression)	\$10,444
TMB/BRAF-guided	Nivolumab Cost	\$18,559	Utility (No Progression)	\$17,584	Utility (Progression)	\$12,559

The deterministic sensitivity analysis evaluates the impact of parameter uncertainty on the base case results through one-way sensitivity analysis. Tornado diagrams illustrate the parameters with the greatest influence on the incremental net monetary benefit.

4.2.2 Probabilistic Sensitivity Analysis

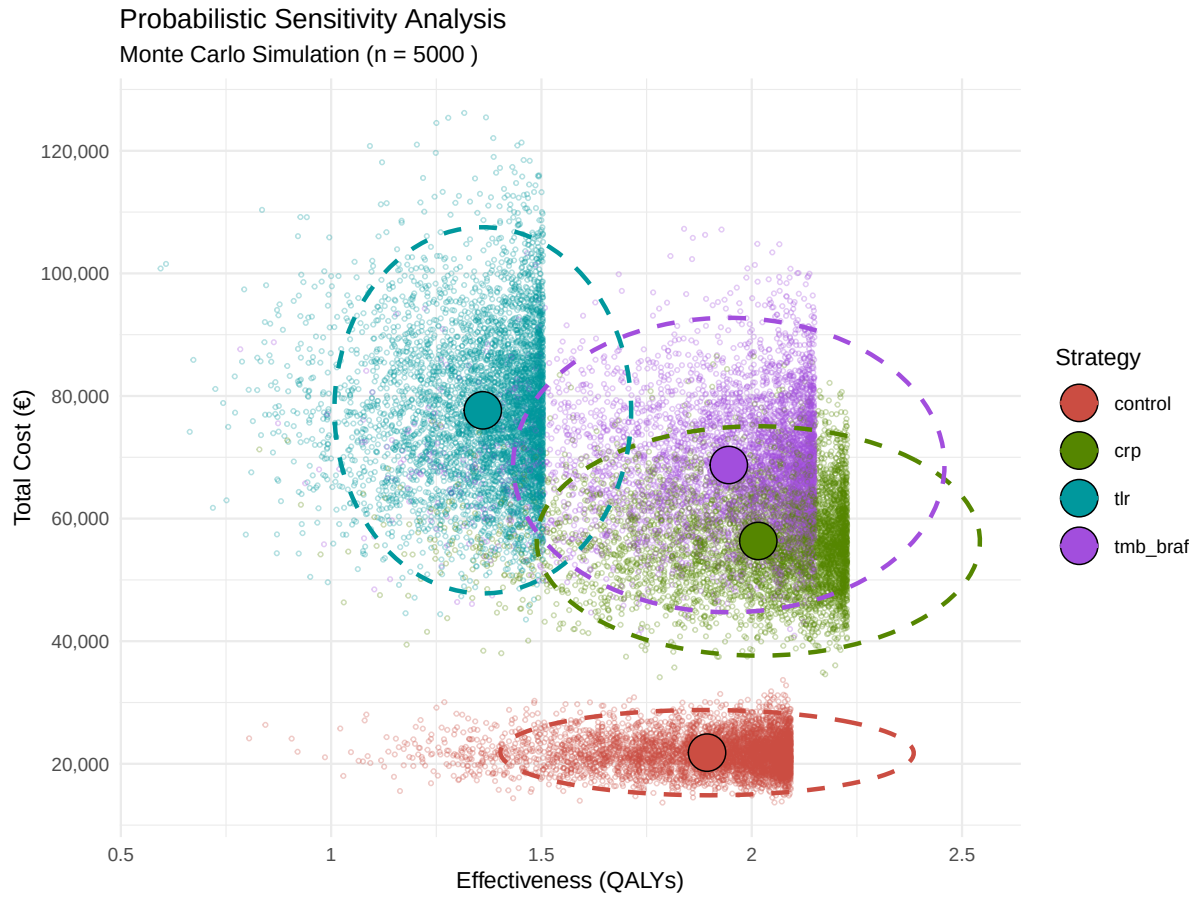


Figure 3: Probabilistic Sensitivity Analysis Scatterplot

The probabilistic sensitivity analysis incorporates parameter uncertainty through Monte Carlo simulation, providing a comprehensive assessment of decision uncertainty across the full range of plausible parameter values.

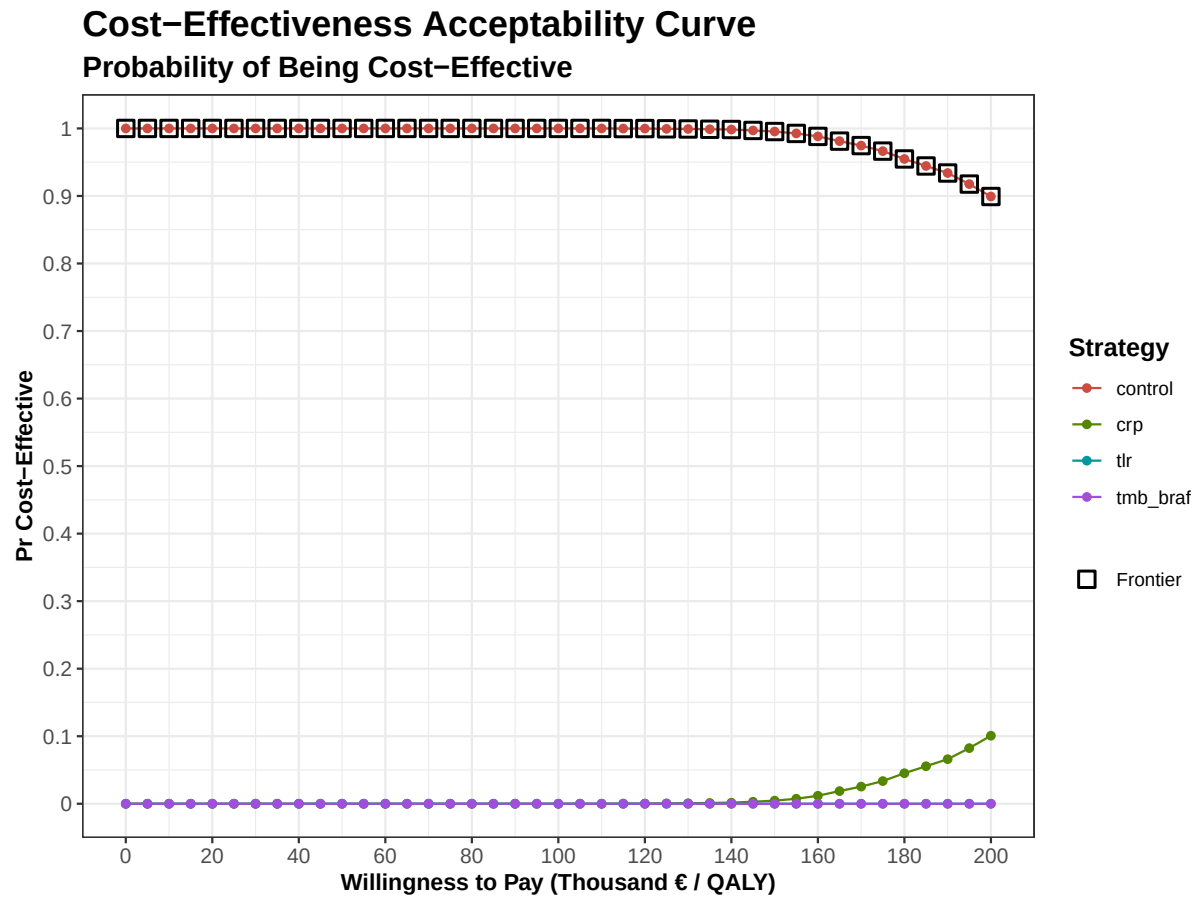


Figure 4: Cost-Effectiveness Acceptability Curve

The cost-effectiveness acceptability curve shows the probability that each strategy is cost-effective across a range of willingness-to-pay thresholds, providing decision-makers with information about uncertainty in the optimal treatment choice.

5 Discussion

5.1 Key Findings

****Analysis Completion Summary****

Result objects available: 15

Analysis matrices available: 0

Base case analysis completed

Deterministic sensitivity analysis completed (15 scenarios)

Optimal strategy identified: control

Analysis conducted at WTP threshold: € 51,000

The cost-effectiveness analysis provides comprehensive evidence for decision-making regarding biomarker-guided treatment strategies in metastatic MSS/pMMR colorectal cancer. The analysis demonstrates the relative value of different biomarker approaches considering both clinical effectiveness and economic efficiency.

5.2 Clinical Implications

The results inform clinical decision-making regarding optimal biomarker strategies for patient selection based on biomarker prevalence and treatment efficacy in the target population. The analysis accounts for the population-level impact of different biomarker prevalences, which is crucial for real-world implementation.

5.3 Economic Implications

From a healthcare payer perspective, the analysis demonstrates value-for-money considerations for different biomarker approaches and their population-level impact on healthcare resource allocation. The sensitivity analyses provide important insights into parameter uncertainty and decision robustness.

5.4 Limitations

This analysis has several limitations including model assumptions regarding long-term outcomes, uncertainty in biomarker performance characteristics, and generalizability to different healthcare systems and populations. The model structure assumes perfect biomarker testing and treatment adherence, which may not reflect real-world clinical practice.

6 Conclusions

The cost-effectiveness analysis provides evidence-based insights for clinical decision-making and healthcare policy regarding biomarker-guided treatment strategies for metastatic MSS/pMMR colorectal cancer. The comprehensive uncertainty analysis supports robust decision-making under uncertainty, while the population-level perspective ensures relevance for healthcare system planning and resource allocation.

The analysis demonstrates the importance of considering biomarker prevalence in addition to test performance characteristics when evaluating population-level cost-effectiveness of precision medicine approaches. Future research should focus on validation of biomarker performance in broader patient populations and assessment of implementation considerations in routine clinical practice.

Analysis completed on: 2025-10-06
Repository: ben-geisler/METIMMOX-1
Analysis version: 1.0
WTP threshold: €51,000